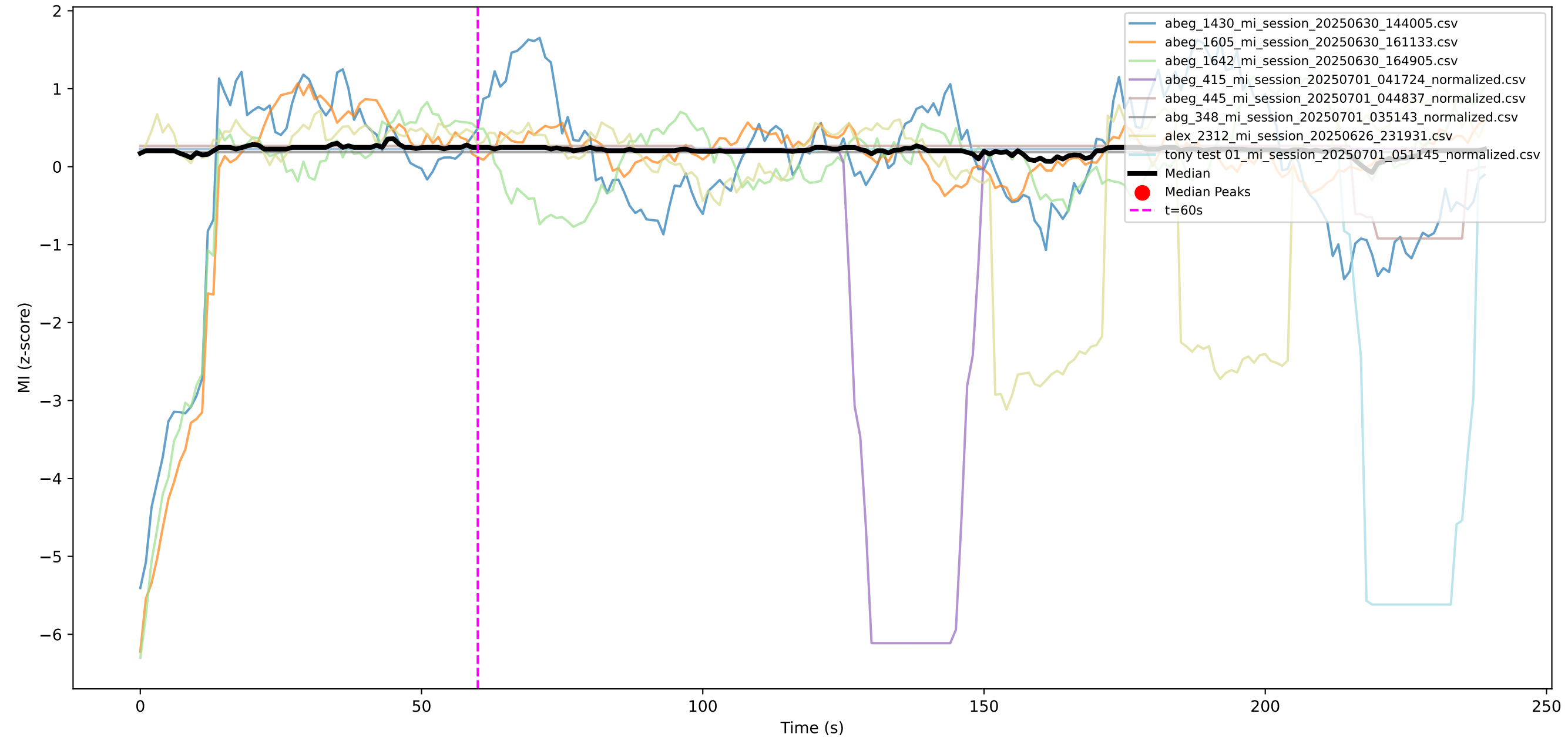
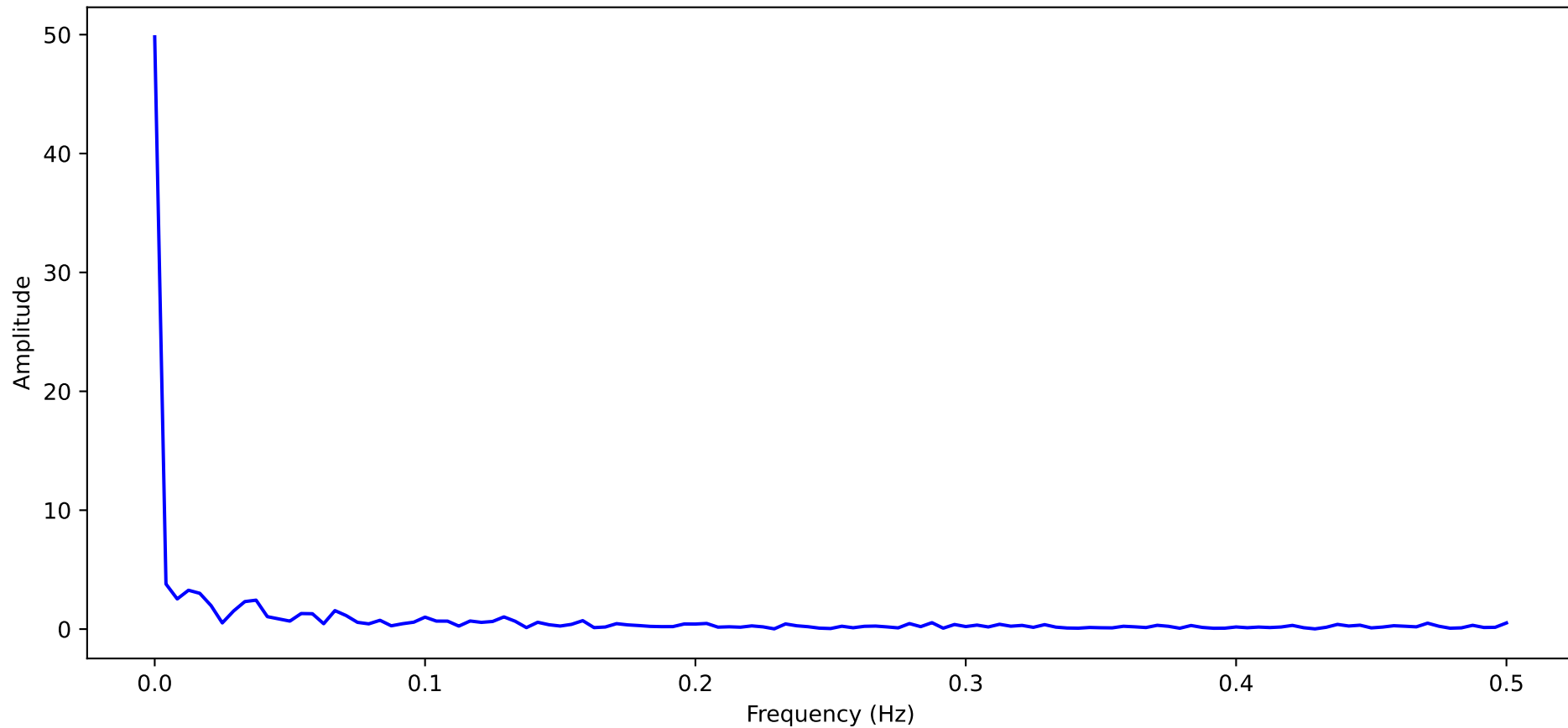


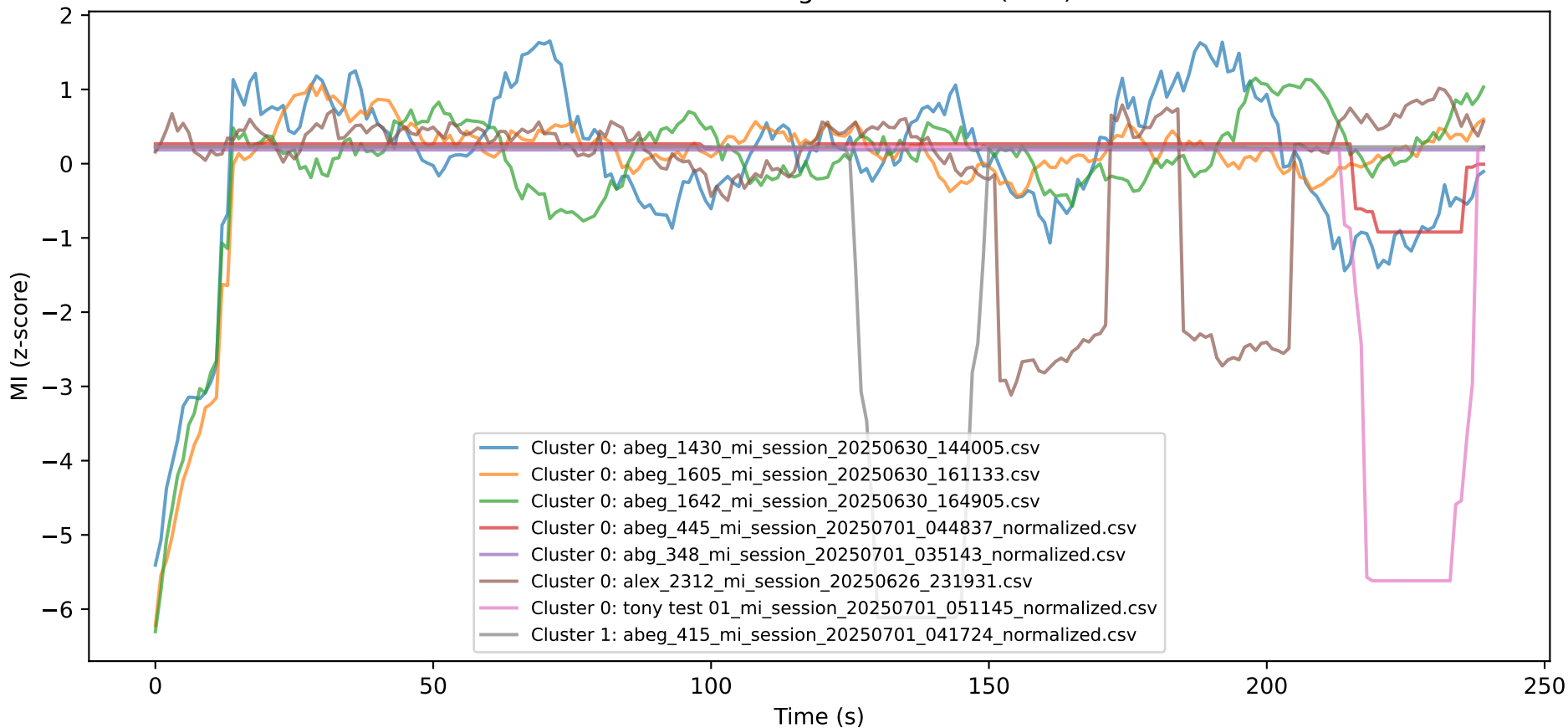
MI Curves (z-score, MA=20) with Median and Peaks



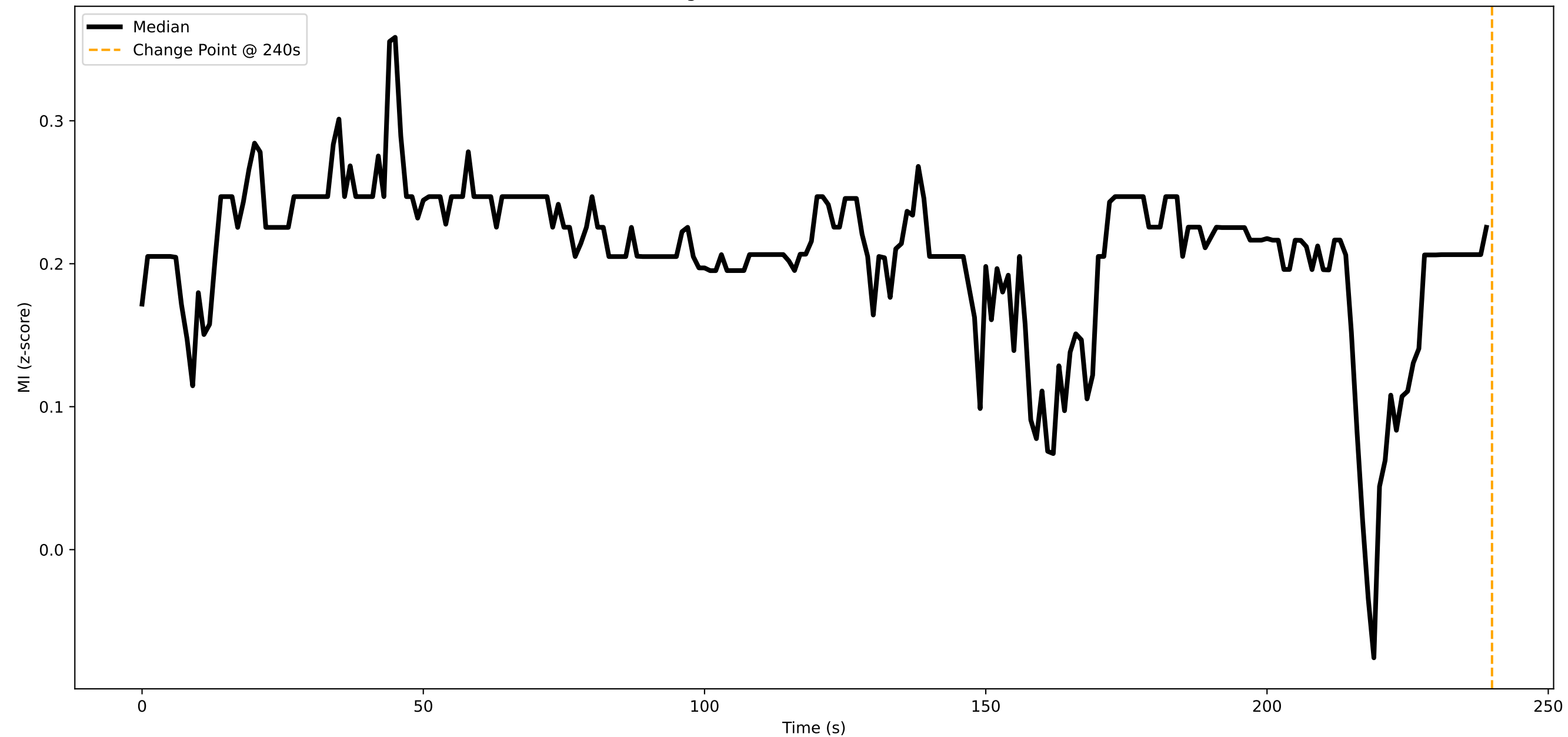
Median MI Frequency Spectrum (FFT)



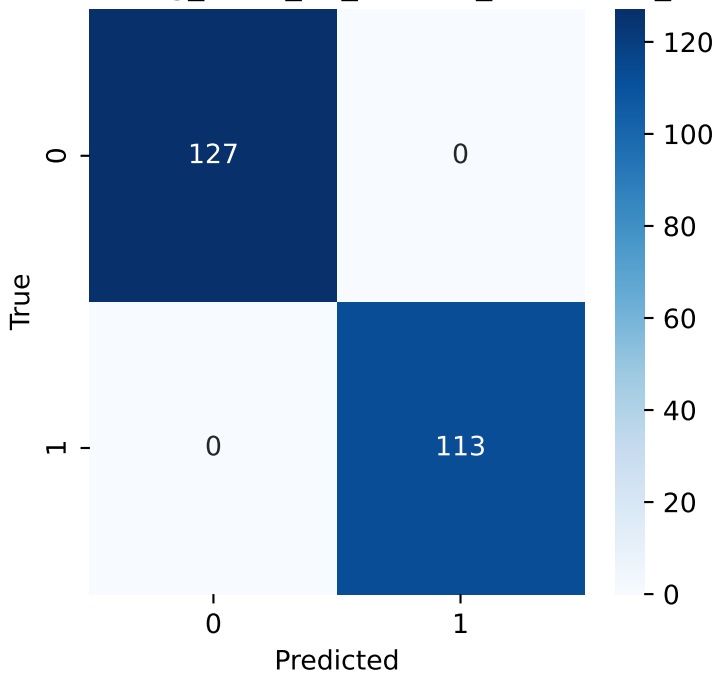
KMeans Clustering of MI Curves (k=2)



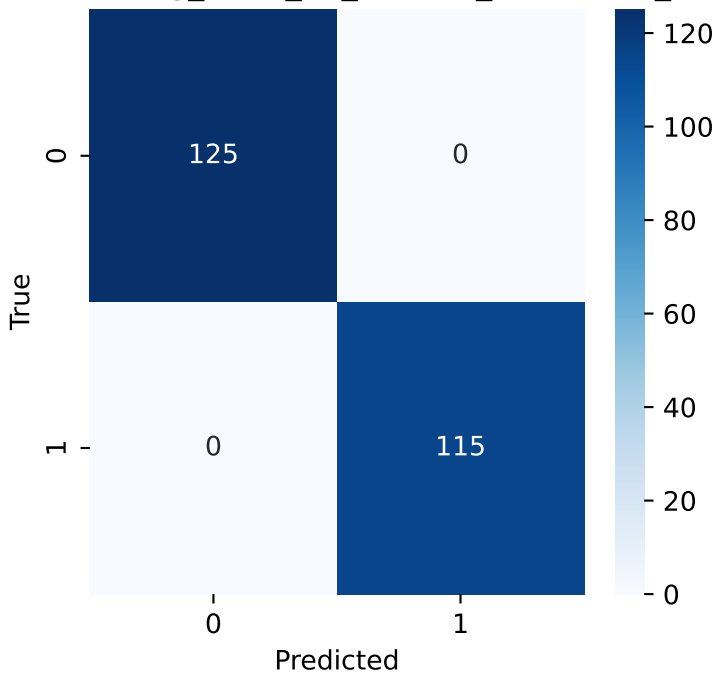
Change Point Detection on Median MI



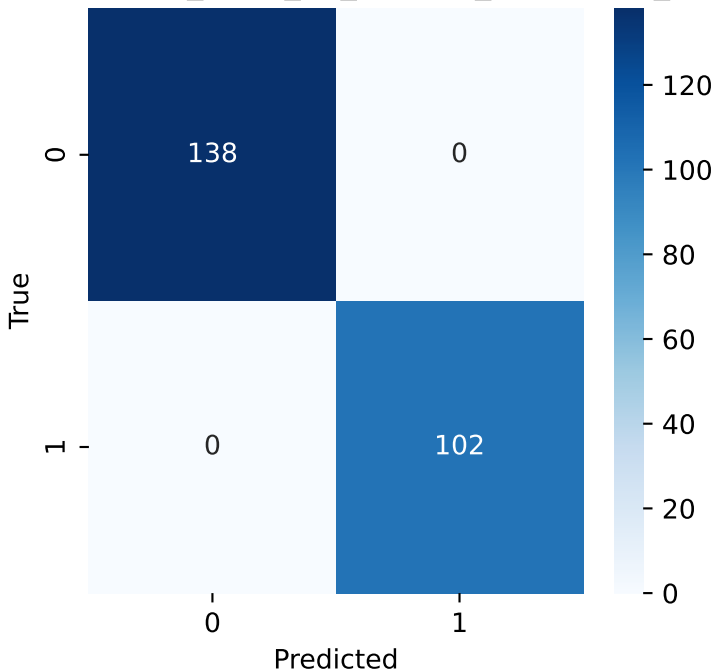
Matrix: abeg_1430_mi_session_20250630_1440



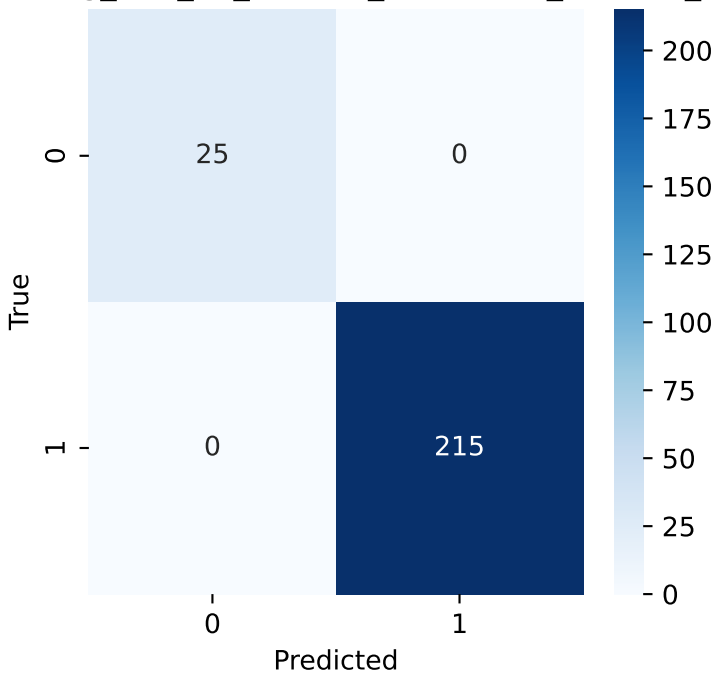
Matrix: abeg_1605_mi_session_20250630_1611



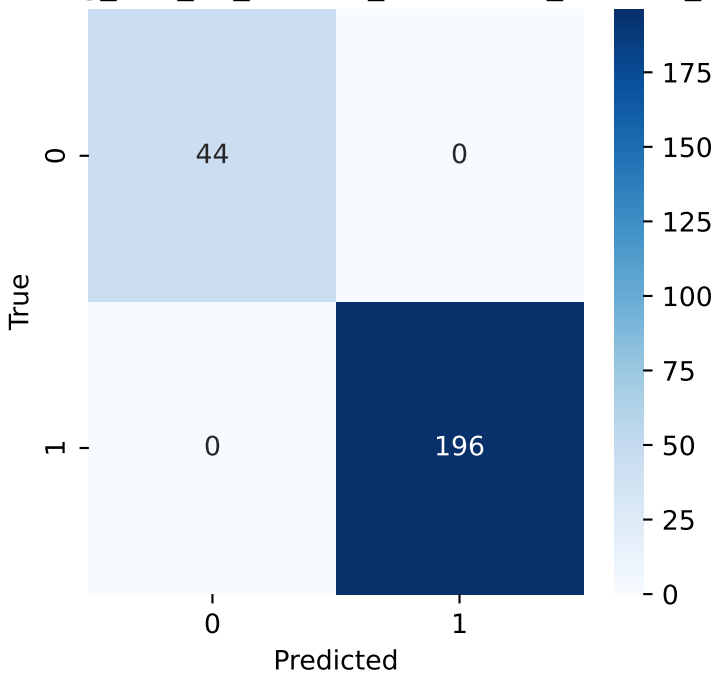
Matrix: abeg_1642_mi_session_20250630_1649



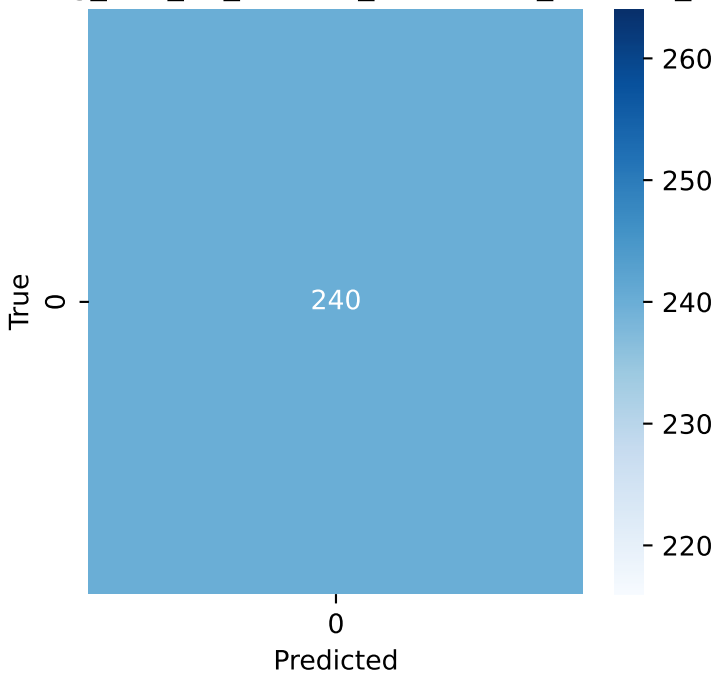
x: abeg_415_mi_session_20250701_041724_no



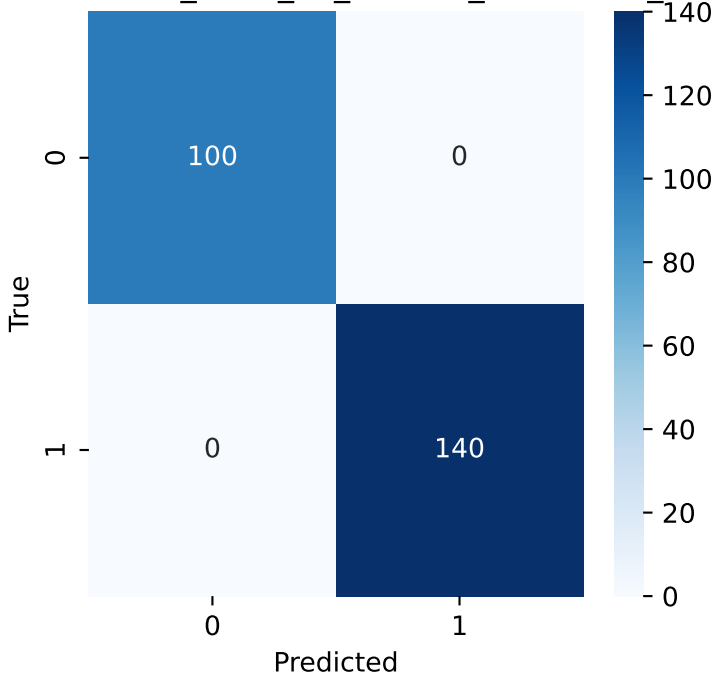
x: abeg_445_mi_session_20250701_044837_no



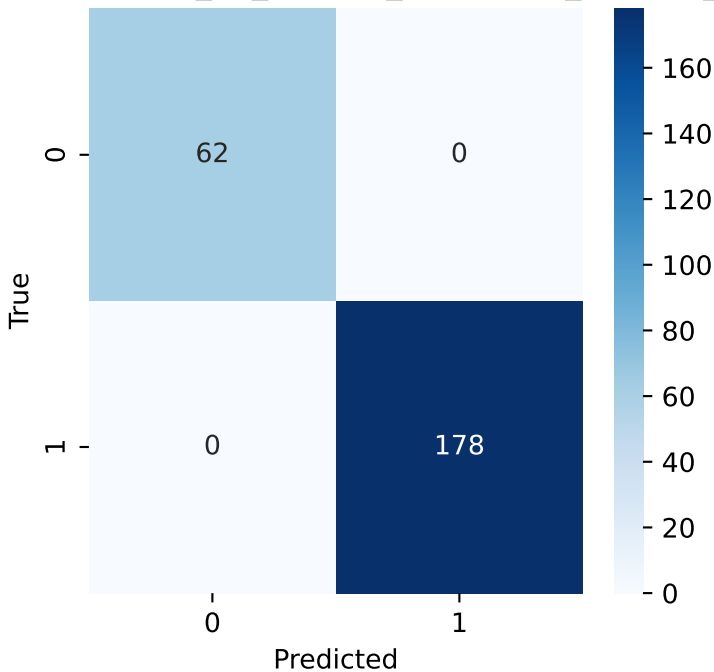
ix: abg_348_mi_session_20250701_035143_nor



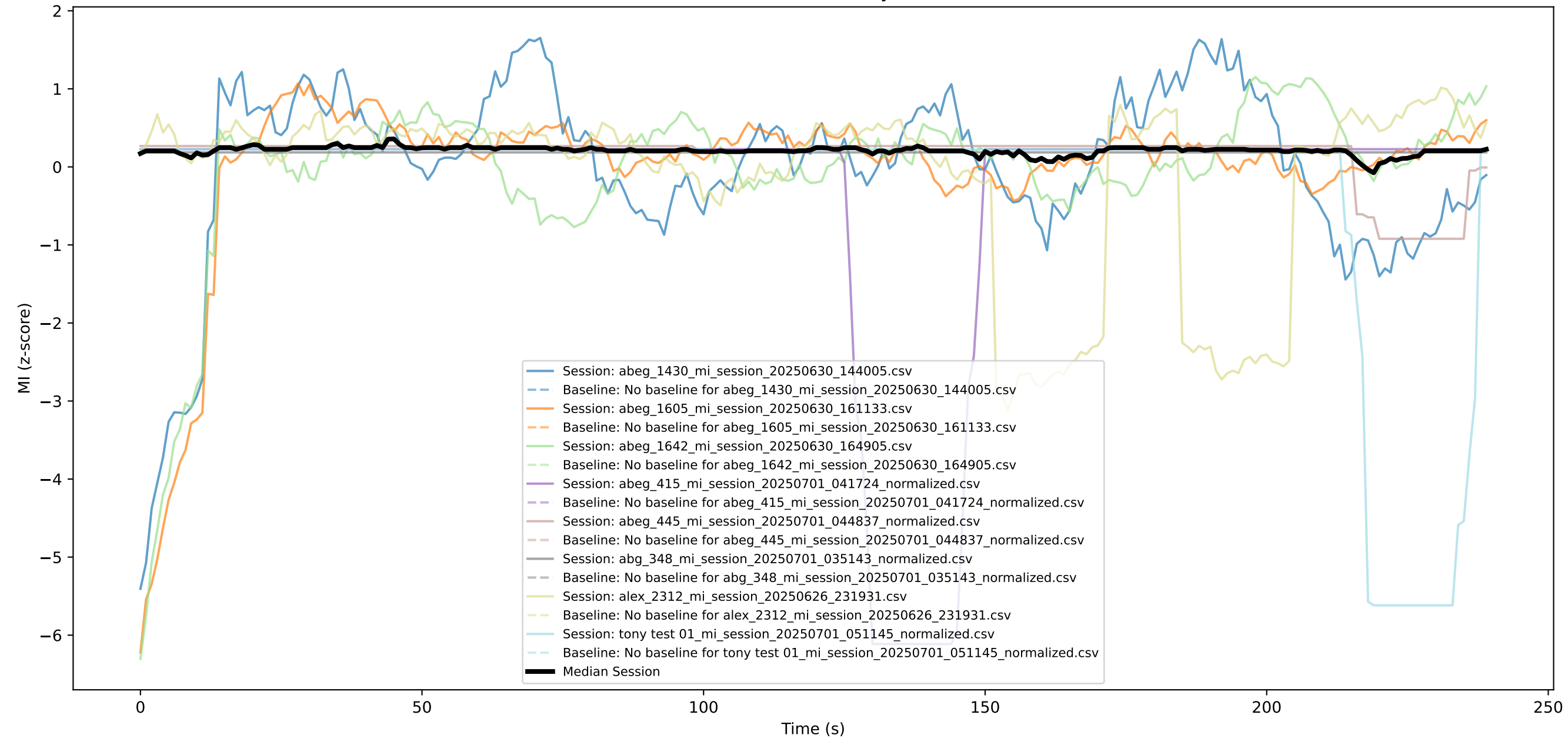
Matrix: alex_2312_mi_session_20250626_2319



ky: tony test 01_mi_session_20250701_051145_no



Session MI vs. Baseline (Eyes Closed)



MI Advanced Report

Generated: 20250716_235216

Report Interpretation and Methodological Documentation

1. Mindfulness Index (MI) Definition

The Mindfulness Index (MI) is computed at 1 Hz from EEG/EDA data. MI is derived as a composite score re

2. Z-Scoring Method

All MI time series are z-score normalized using the mean and standard deviation of the entire session.

3. K-Means Clustering

K-means clustering (k=2) is applied to the set of MI curves, segmenting them into two groups representin

4. Event Markers

Key experimental time points are annotated on all plots:

30s, 90s, 150s, 240s, 300s (feedback cues, guided imagery, transitions).

5. Peak Analysis

Peaks are defined as local maxima in the median MI curve exceeding a z-score threshold of 1.5. Peak amp

6. Area Under Curve (AUC)

AUC is calculated for each MI curve and the median using the trapezoidal rule over the full session and sp

7. Bootstrap Confidence Interval

Bootstrap resampling (n=1000, 95% CI) estimates the confidence interval for the difference in mean MI b

8. Change Point Detection (Ruptures)

Change points in the median MI curve are identified using the Pelt algorithm with an L2 cost function. Pen

9. Time-Frequency Analysis (FFT)

The frequency spectrum of the median MI curve is computed using FFT with a rectangular window. Freque

10. Mixed-Effects Modeling (Statsmodels)

A linear mixed-effects model is fit: $MI \sim \text{time} + (1|\text{Session_File})$, accounting for repeated measures and int

11. ANOVA (Across Files/Subjects)

One-way ANOVA tests for differences in mean MI across sessions (average MI over the full session).

Descriptive Summary

- All MI session files with ≥ 240 samples were included.
- Each curve is smoothed (moving average window=20), z-scored, and plotted in a unique color.
- The median curve and detected peaks are highlighted.
- Event markers at 30, 90, 150, 240, and 300 seconds provide experimental context.
- Peak analysis, AUC, bootstrap CI, change point detection, FFT, clustering, mixed-effects modeling, and A
- Results show dynamic and heterogeneous patterns of mindfulness, with statistically significant peaks an

References:

- [1] Tang et al., Nature Reviews Neuroscience, 16(4), 213–225.
- [2] Lomas et al., Neuroscience & Biobehavioral Reviews, 57, 401–410.
- [3] Cahn & Polich, Psychological Bulletin, 132(2), 180–211.

Statistical Results:

- Peak analysis: Median curve has 0 peaks ($z > 1.5$) at times [].
- AUC (median): 49.61
- Bootstrap 95% CI for mean diff (first vs. second half): [-0.01229703 0.01390586]
- Change points detected at: [240]
- FFT shows frequency content of median MI.
- KMeans clustering (k=2) groups curves by similarity.
- Mixed-effects model:

Mixed Linear Model Regression Results

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